

Advanced Image Processing Methods

Homework 5 Report

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Goal

Build a complete PyTorch transfer-learning image-classification pipeline: load a `torchvision` dataset, fine-tune a pretrained model, validate with early stopping, then evaluate the best checkpoint on a held-out test set.

Method

1. **Data preparation:** The Describable Textures Dataset (DTD) is loaded through `torchvision`, using official `partition=1`: 1880 train, 1880 validation, and 1880 test images across 47 texture classes. Training images are resized to 224×224 and augmented online with random horizontal flip, rotation, and colour jitter; validation and test images use only resize, tensor conversion, and ImageNet normalisation.
2. **Model:** The backbone is MaxViT-T with ImageNet pretrained weights. The final classifier layer is replaced by a 47-way `Linear` layer, and all 30.43M parameters remain trainable for full fine-tuning.
3. **Training:** The model is trained with `CrossEntropyLoss` and Adam. After every epoch, validation is run under `model.eval()` and `torch.no_grad()`. Early stopping monitors validation loss and saves only the best checkpoint.
4. **Evaluation:** The best checkpoint is reloaded before test evaluation. The script reports test loss and accuracy, saves train/validation curves, and plots a 47-class confusion matrix for test predictions.

The main hyperparameters were: image size 224×224 , CUDA batch size 32, Adam learning rate 10^{-4} , weight decay 10^{-4} , 30 epoch budget, and early-stopping patience 5. The final run used Kaggle's Tesla T4 environment with PyTorch 2.10.0+cu128.

Results

Training stopped after epoch 14. The best validation checkpoint was reached at epoch 9 with validation loss 1.0349 and validation accuracy 0.7234. On the test split, the reloaded best model achieved **72.71% accuracy** with test loss **1.0189**. The curves show rapid transfer-learning convergence followed by mild overfitting: training loss keeps decreasing after epoch 9, while validation loss plateaus and then rises slightly.

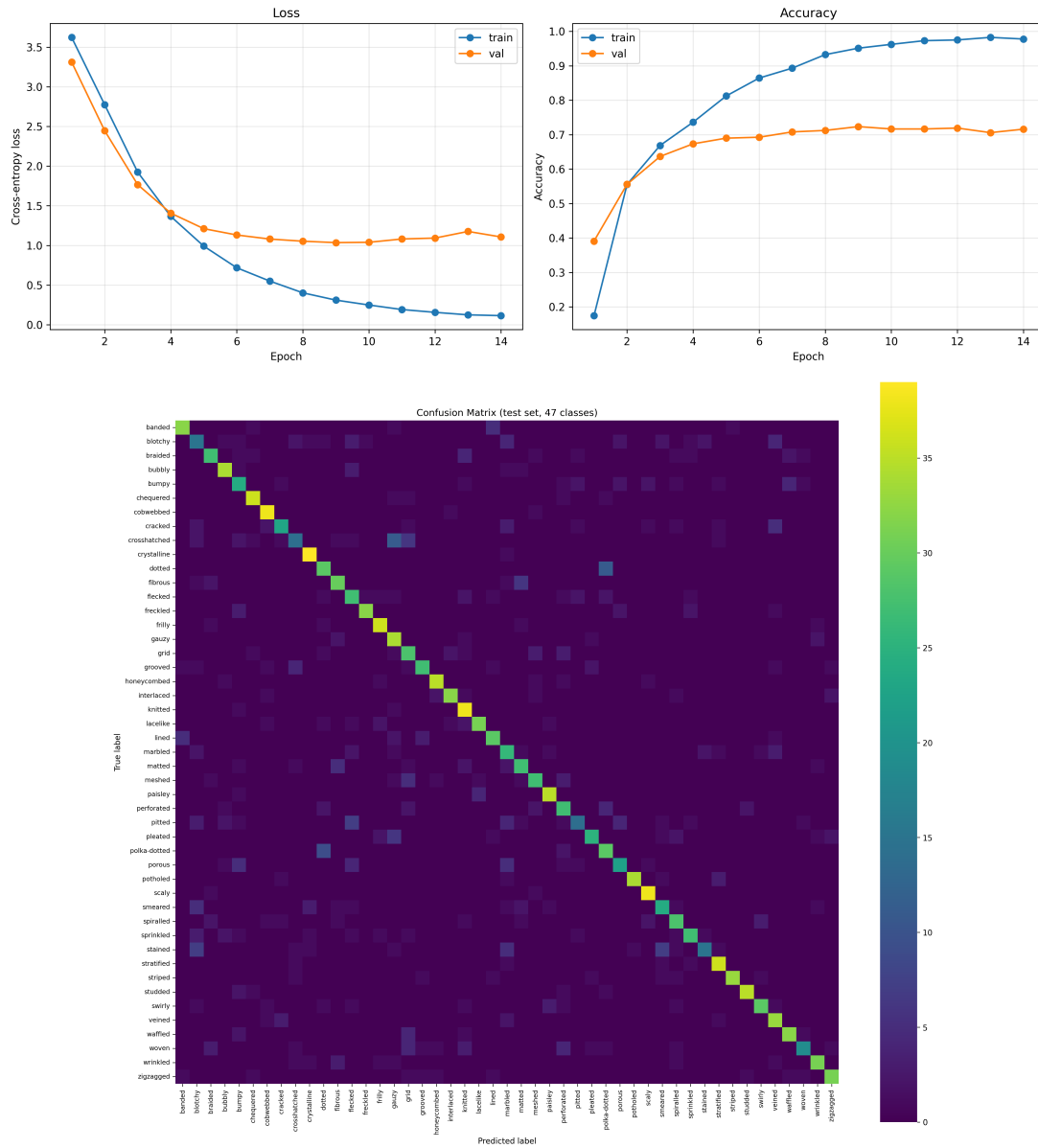


Figure 1: Training/validation curves and test-set confusion matrix. The strong diagonal confirms good overall classification; most mistakes are scattered among visually related texture categories such as dotted/polka-dotted, grid/lined/crosshatched, and fibrous/matted/woven patterns.